

optimizing logistic regression

CSCI
270

logistic regression

model space

$$y = \left(\begin{bmatrix} \theta_1 \\ \vdots \\ \theta_D \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_D \end{bmatrix} > 0 \right)$$

loss function

$$L(\theta) = - \sum_{n=1}^N \log(p^{(n)} y^{(n)} + (1-p^{(n)})(1-y^{(n)}))$$

$$\text{where } p^{(n)} = \sigma \left(\begin{bmatrix} \theta_1 \\ \vdots \\ \theta_D \end{bmatrix} \cdot \begin{bmatrix} x_1^{(n)} \\ \vdots \\ x_D^{(n)} \end{bmatrix} \right)$$

training data

		instances		
		1	...	N
features	x_1	$x_1^{(1)}$	...	$x_1^{(N)}$
	$\vdots$	$\vdots$		$\vdots$
	x_D	$x_D^{(1)}$	...	$x_D^{(N)}$
outcome	y	$y^{(1)}$	...	$y^{(N)}$
		binary values		

optimization

$$\underset{\theta}{\operatorname{argmin}} L(\theta)$$

finally we need the

gradient $\frac{\partial L}{\partial \theta}$



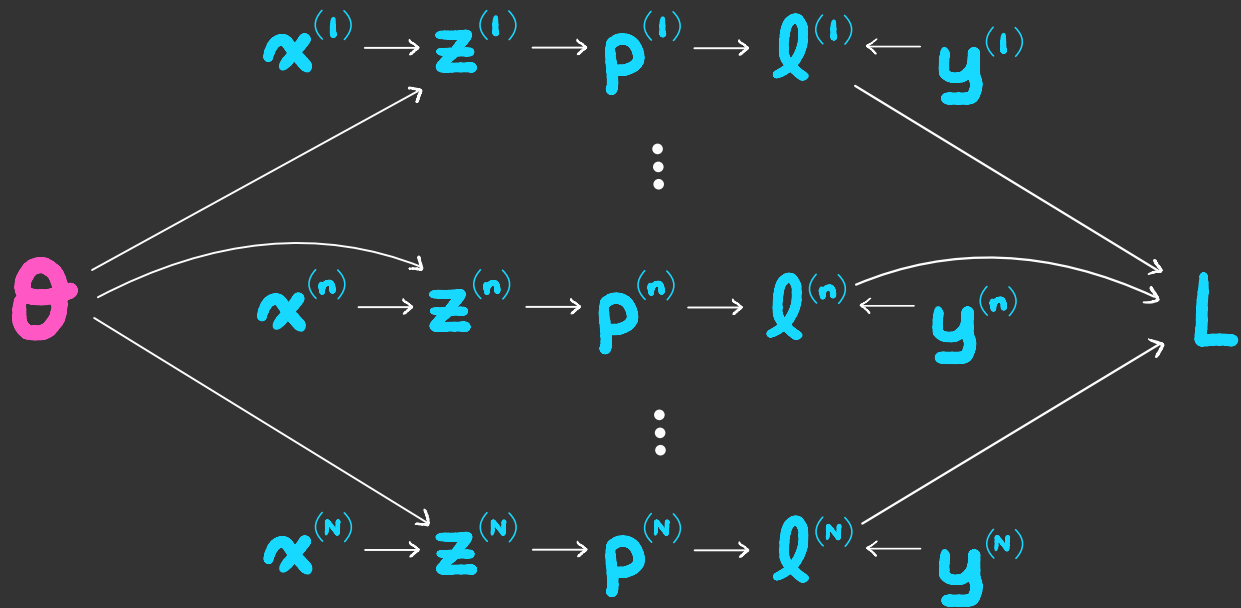
use a function
graph!



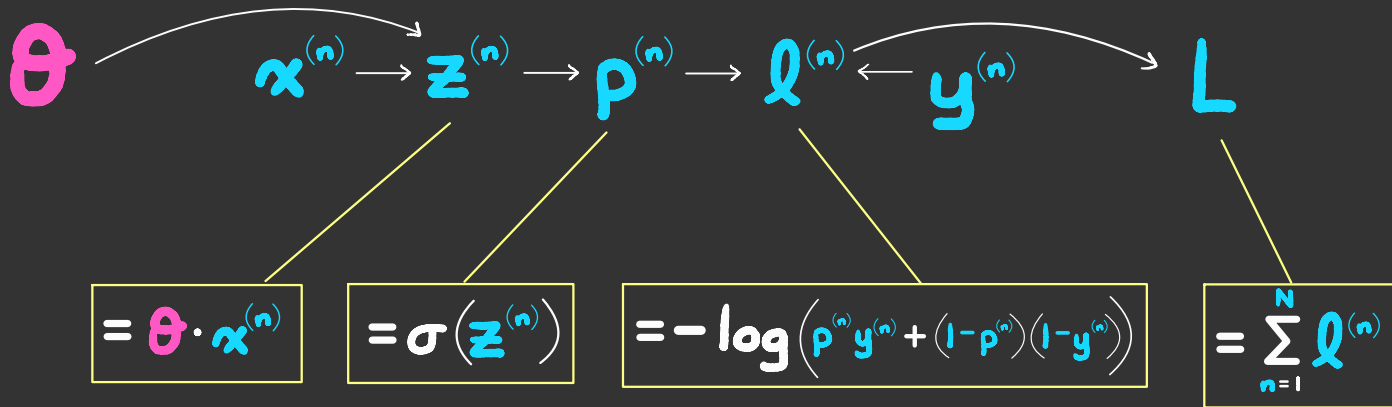
use a function
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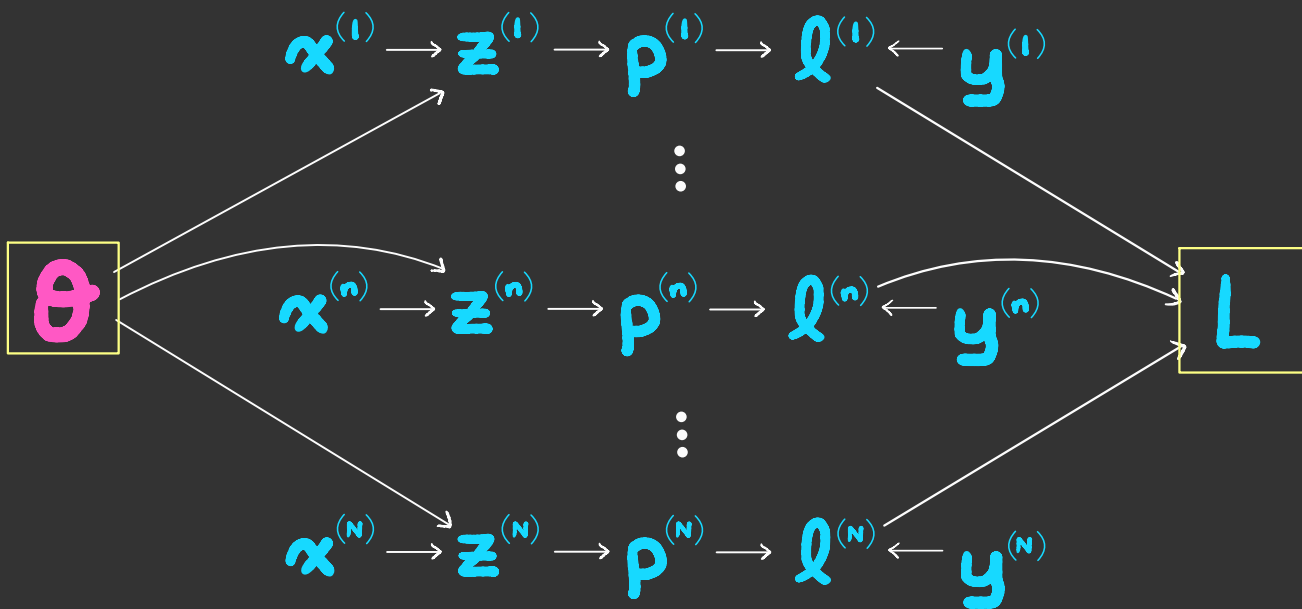
$$L(\theta) = -\sum_{n=1}^N \log(p^{(n)} y^{(n)} + (1-p^{(n)})(1-y^{(n)})) \quad \text{where } p^{(n)} = \sigma\left(\begin{bmatrix} \theta_1 \\ \vdots \\ \theta_D \end{bmatrix} \cdot \begin{bmatrix} x_1^{(n)} \\ \vdots \\ x_D^{(n)} \end{bmatrix}\right)$$



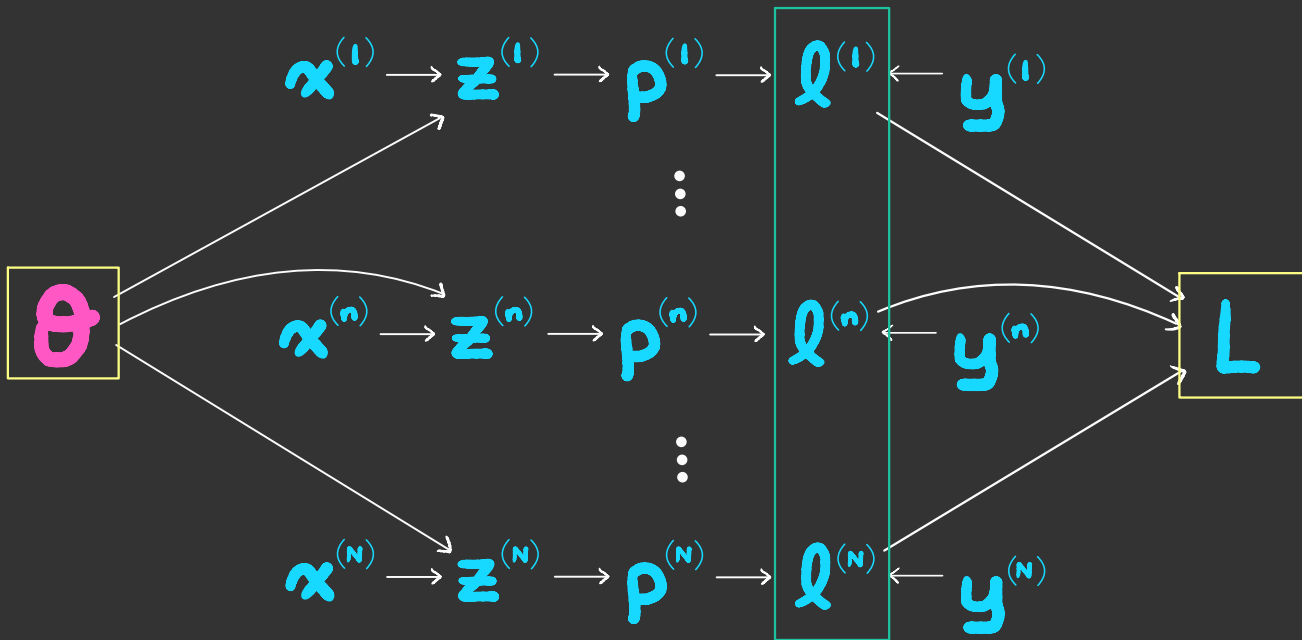
$$L(\theta) = \sum_{n=1}^N -\log(p^{(n)}y^{(n)} + (1-p^{(n)})(1-y^{(n)})) \quad \text{where } p^{(n)} = \sigma\left(\begin{bmatrix} \theta_0 \\ \vdots \\ \theta_D \end{bmatrix} \cdot \begin{bmatrix} x_0^{(n)} \\ \vdots \\ x_D^{(n)} \end{bmatrix}\right)$$



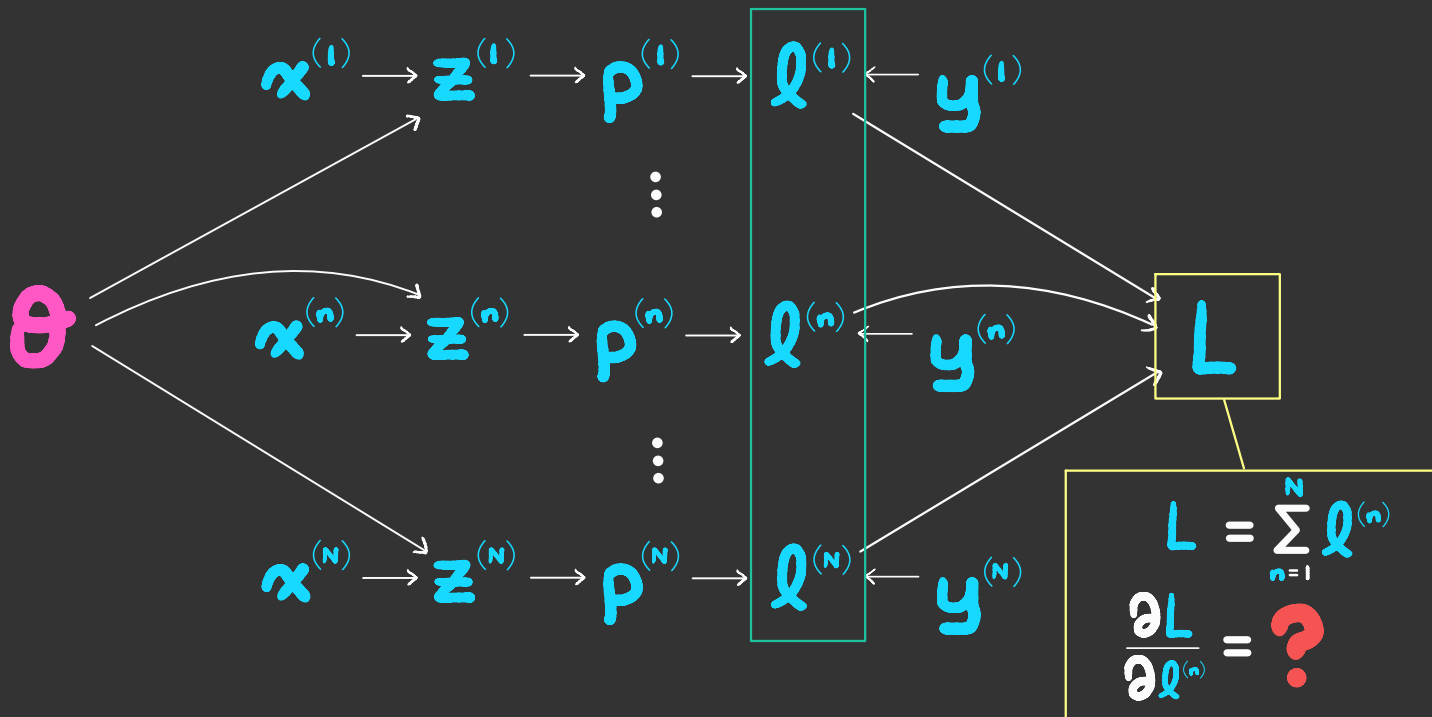
$$\frac{\partial L}{\partial \theta} = ?$$



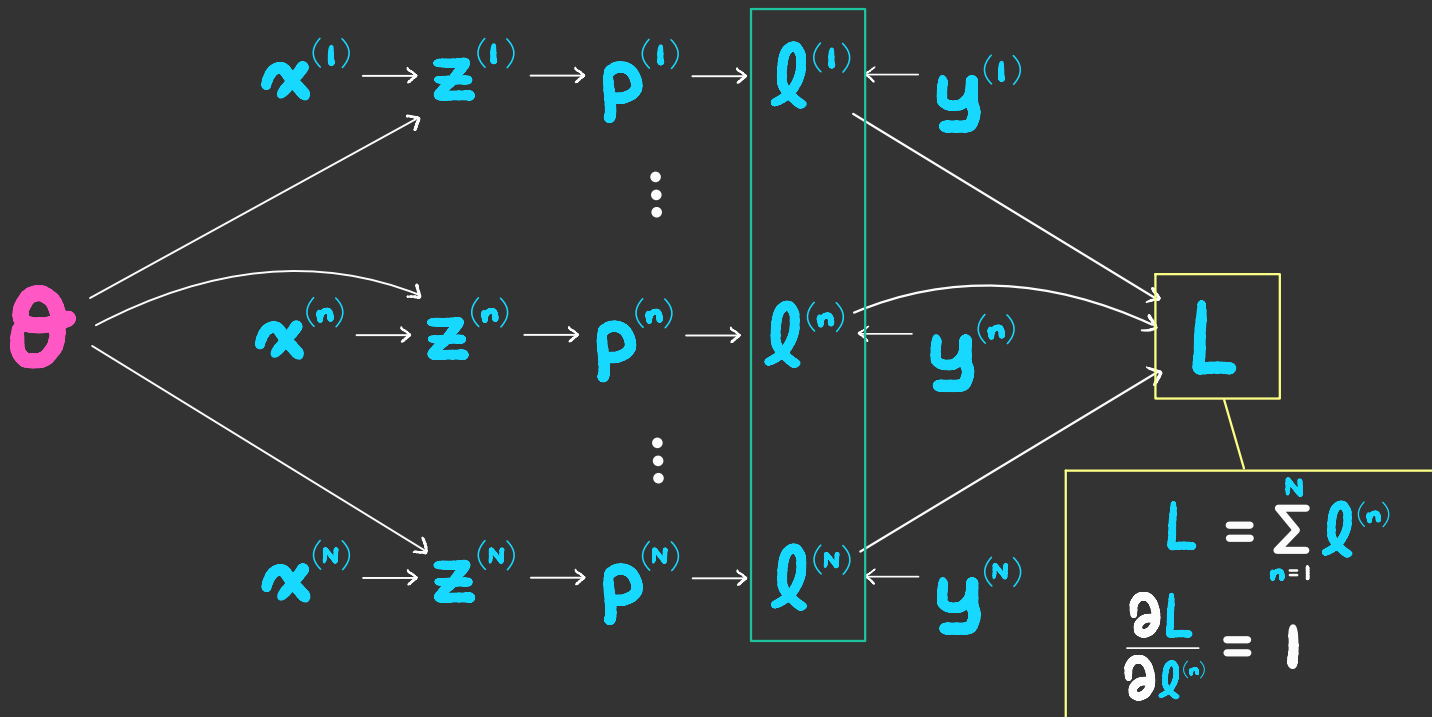
$$\frac{\partial L}{\partial \theta} = \sum_{n=1}^N \frac{\partial L}{\partial q^{(n)}} \frac{\partial q^{(n)}}{\partial \theta}$$



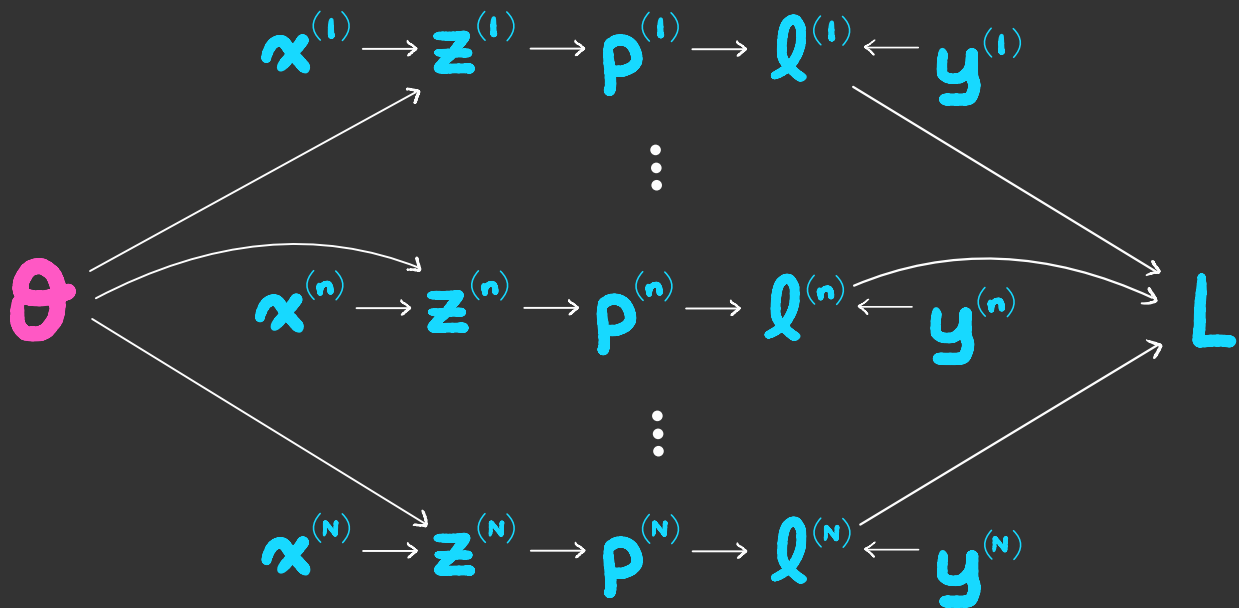
$$\frac{\partial L}{\partial \theta} = \sum_{n=1}^N \boxed{\frac{\partial L}{\partial l^{(n)}}} \frac{\partial l^{(n)}}{\partial \theta}$$



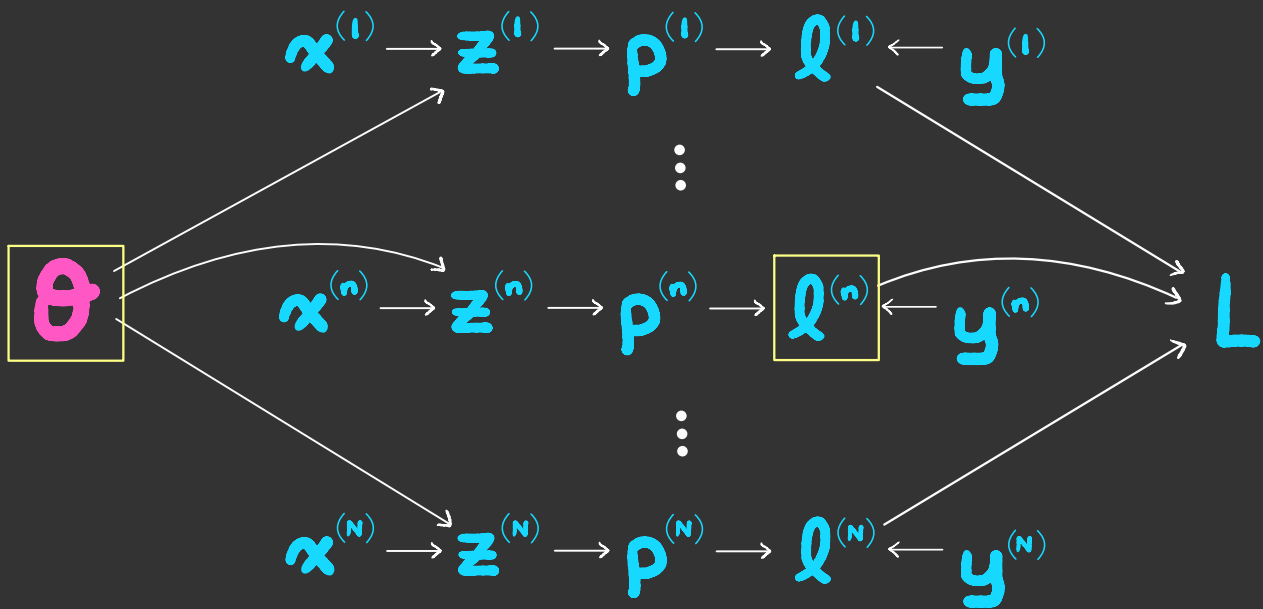
$$\frac{\partial L}{\partial \theta} = \sum_{n=1}^N \boxed{1} \frac{\partial l^{(n)}}{\partial \theta}$$



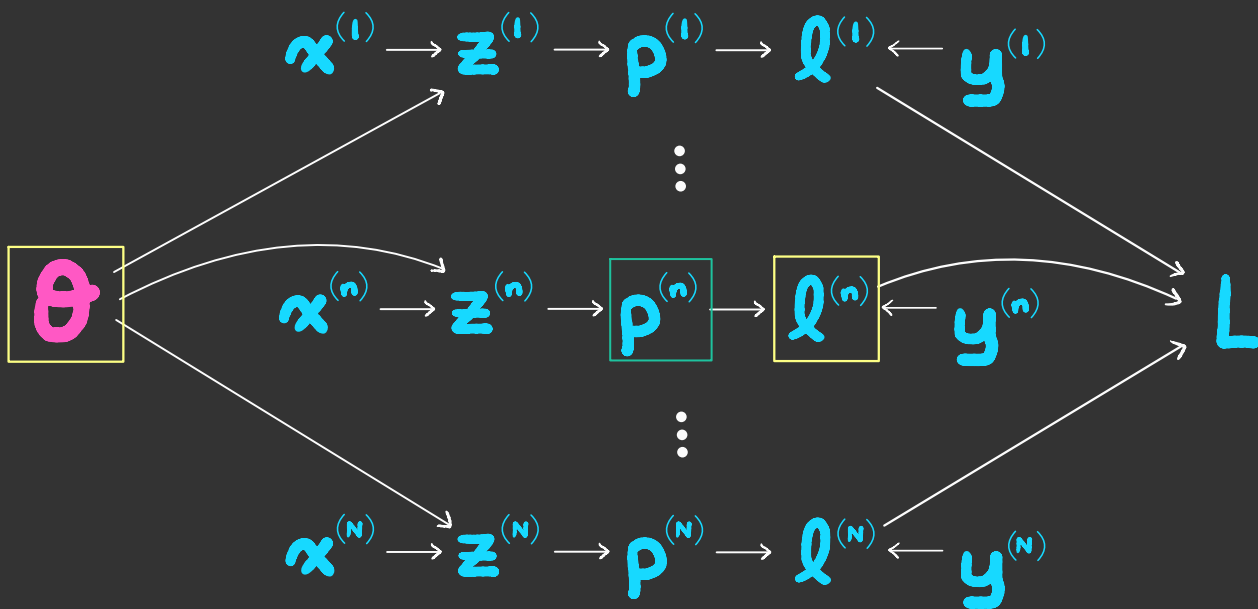
$$\frac{\partial L}{\partial \theta} = \sum_{n=1}^N \frac{\partial l^{(n)}}{\partial \theta}$$



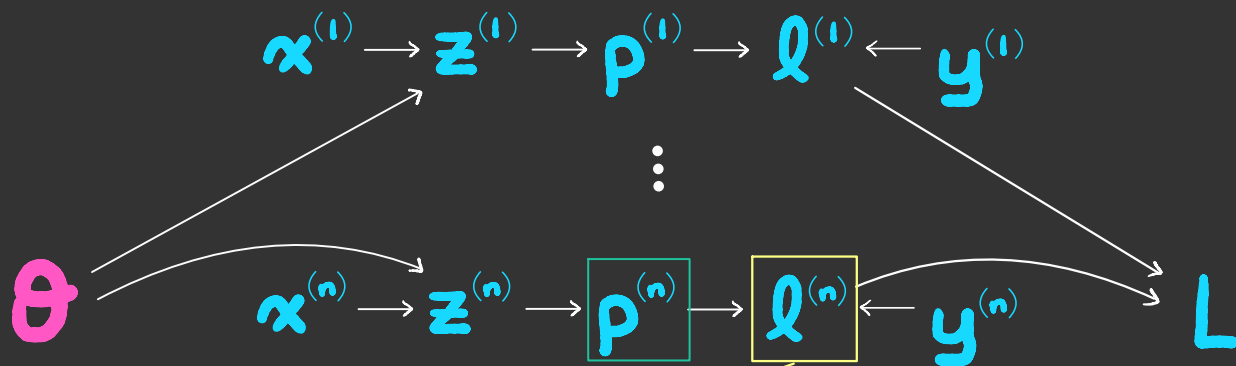
$$\frac{\partial L}{\partial \theta} = \sum_{n=1}^N \boxed{\frac{\partial l^{(n)}}{\partial \theta}}$$



$$\frac{\partial L}{\partial \theta} = \sum_{n=1}^N \frac{\partial l^{(n)}}{\partial p^{(n)}} \frac{\partial p^{(n)}}{\partial \theta}$$



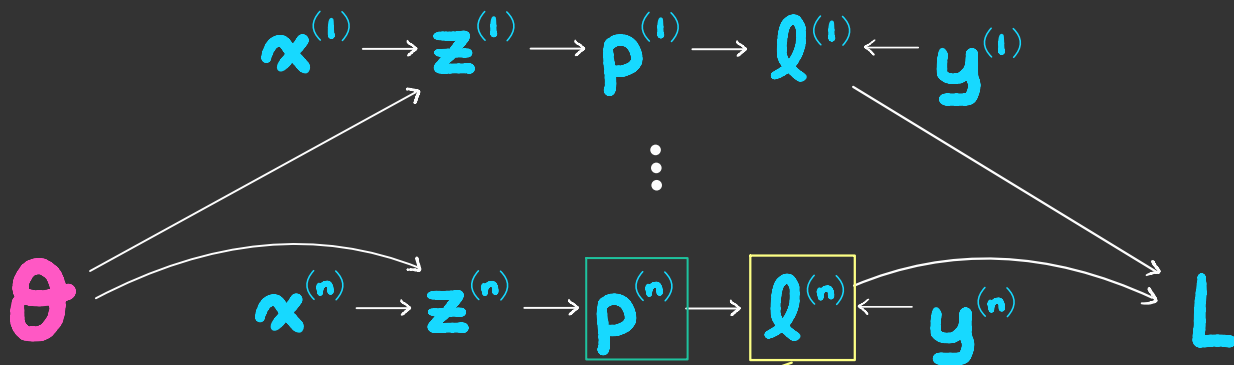
$$\frac{\partial L}{\partial \theta} = \sum_{n=1}^N \boxed{\frac{\partial l^{(n)}}{\partial p^{(n)}}} \frac{\partial p^{(n)}}{\partial \theta}$$



$$l^{(n)} = -\log(p^{(n)}y^{(n)} + (1-p^{(n)})(1-y^{(n)}))$$

$$\frac{\partial l^{(n)}}{\partial p^{(n)}} = ?$$

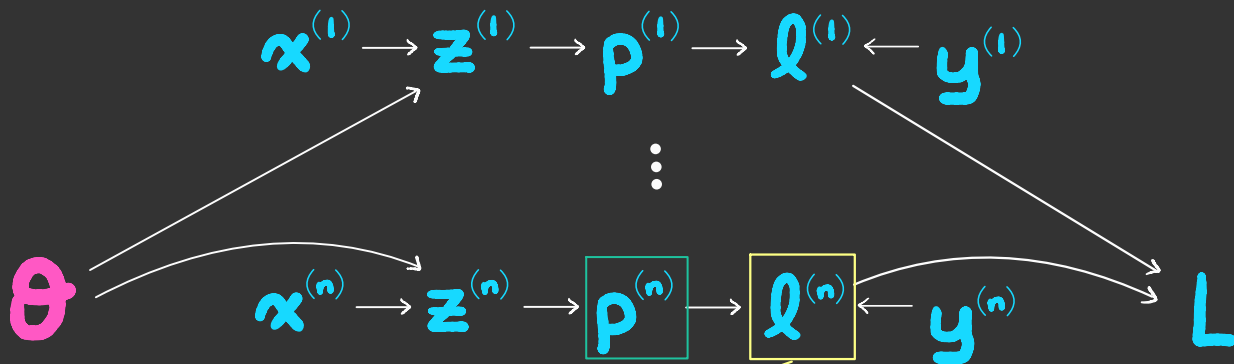
$$\frac{\partial L}{\partial \theta} = \sum_{n=1}^N \boxed{\frac{\partial l^{(n)}}{\partial p^{(n)}}} \frac{\partial p^{(n)}}{\partial \theta}$$



$$l^{(n)} = -\log(p^{(n)}y^{(n)} + (1-p^{(n)})(1-y^{(n)}))$$

$$\frac{\partial l^{(n)}}{\partial p^{(n)}} = \frac{1 - 2y^{(n)}}{p^{(n)}y^{(n)} + (1-p^{(n)})(1-y^{(n)})}$$

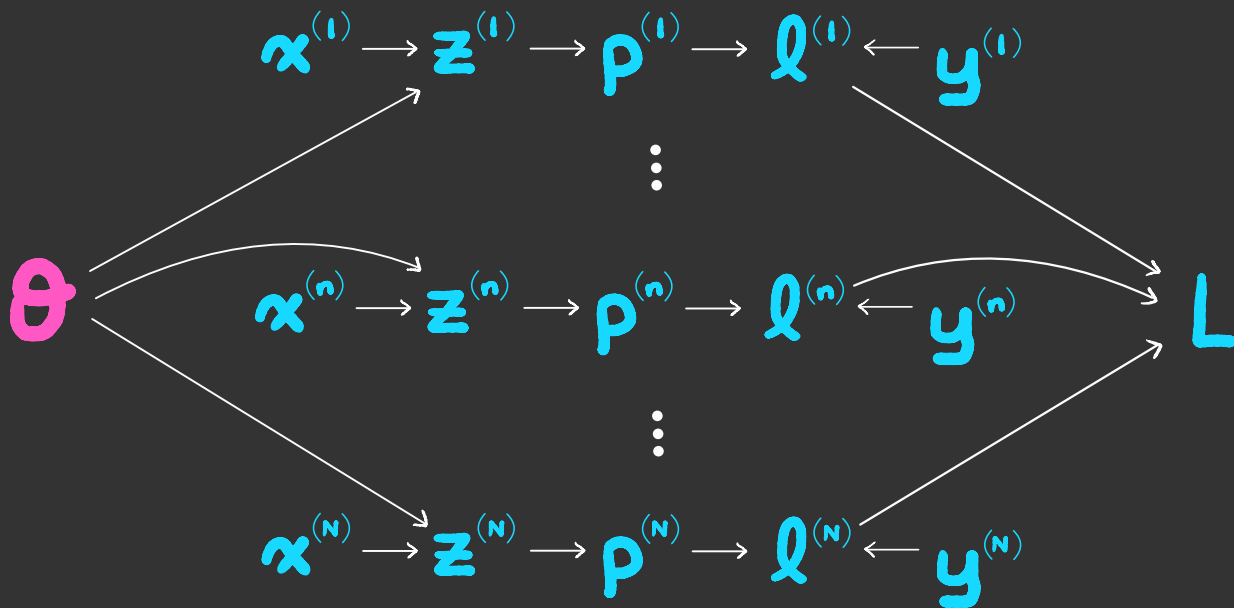
$$\frac{\partial L}{\partial \theta} = \sum_{n=1}^N \left[\frac{1 - 2y^{(n)}}{p^{(n)}y^{(n)} + (1-p^{(n)})(1-y^{(n)})} \right] \frac{\partial p^{(n)}}{\partial \theta}$$



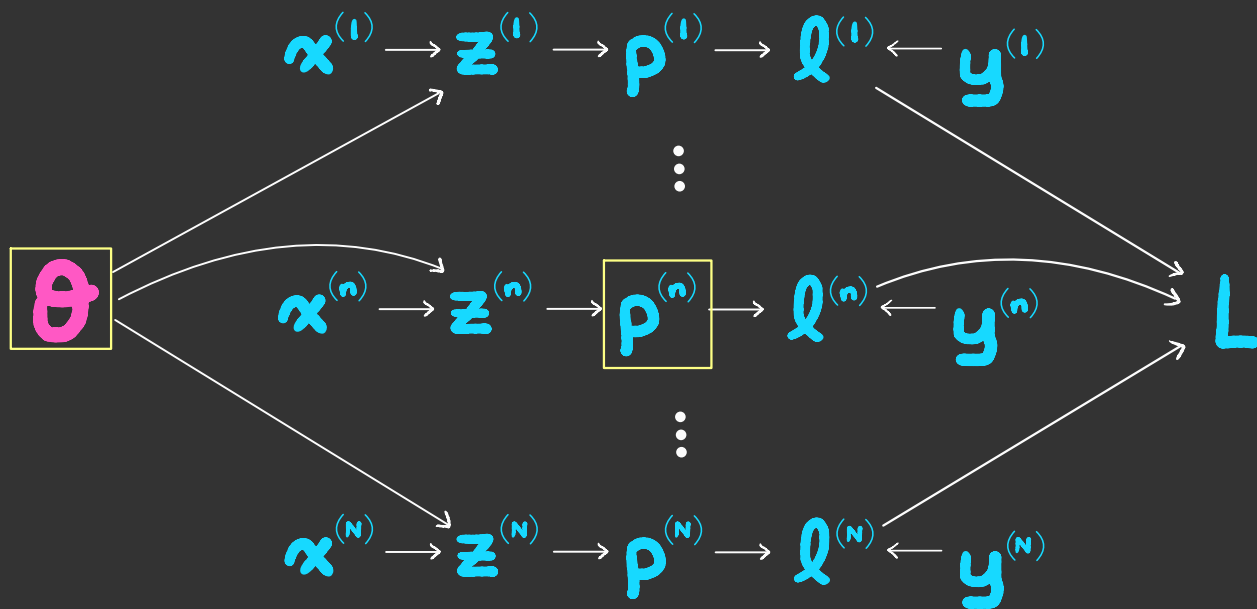
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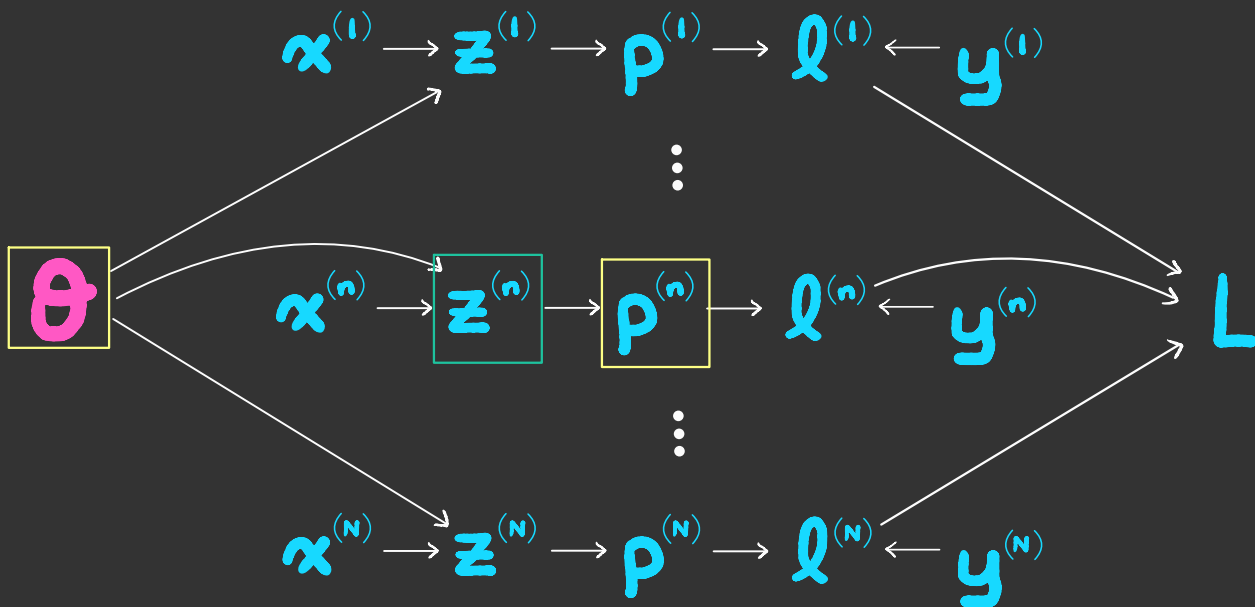
$$\frac{\partial L}{\partial \theta} = \sum_{n=1}^N \frac{1 - 2y^{(n)}}{p^{(n)}y^{(n)} + (1-p^{(n)})(1-y^{(n)})} \frac{\partial p^{(n)}}{\partial \theta}$$



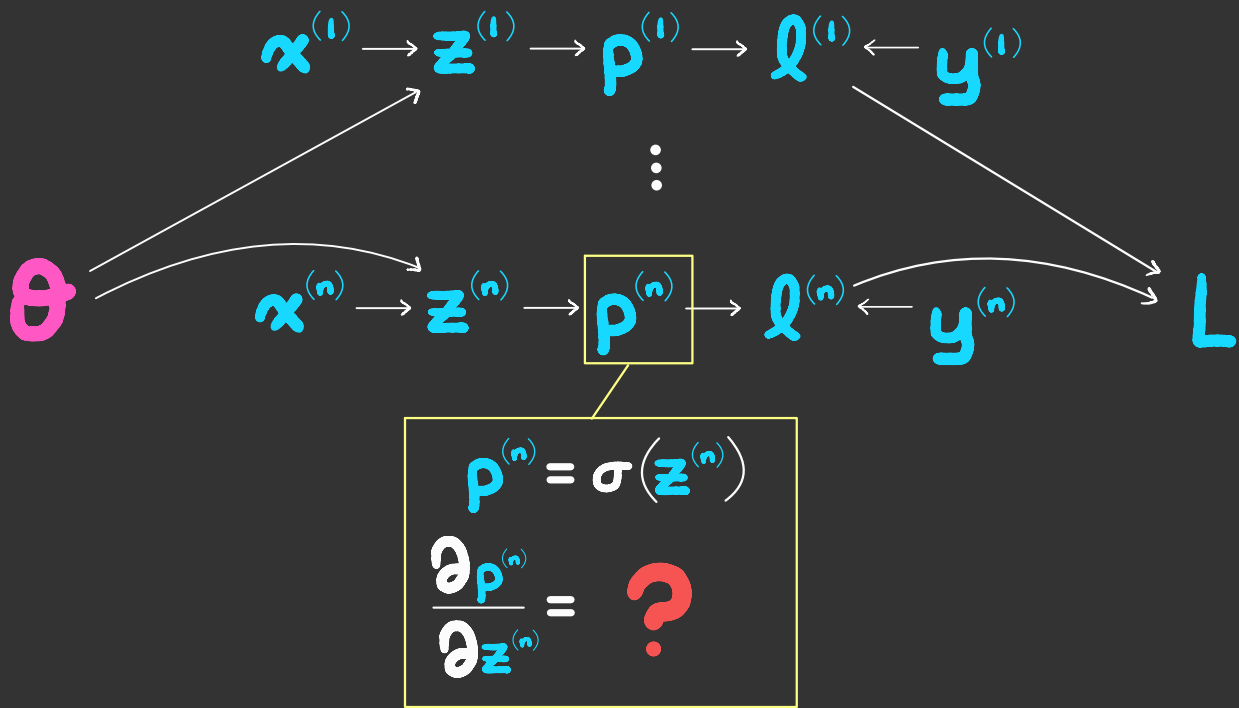
$$\frac{\partial L}{\partial \theta} = \sum_{n=1}^N \frac{1 - 2y^{(n)}}{p^{(n)}y^{(n)} + (1-p^{(n)})(1-y^{(n)})} \boxed{\frac{\partial p^{(n)}}{\partial \theta}}$$



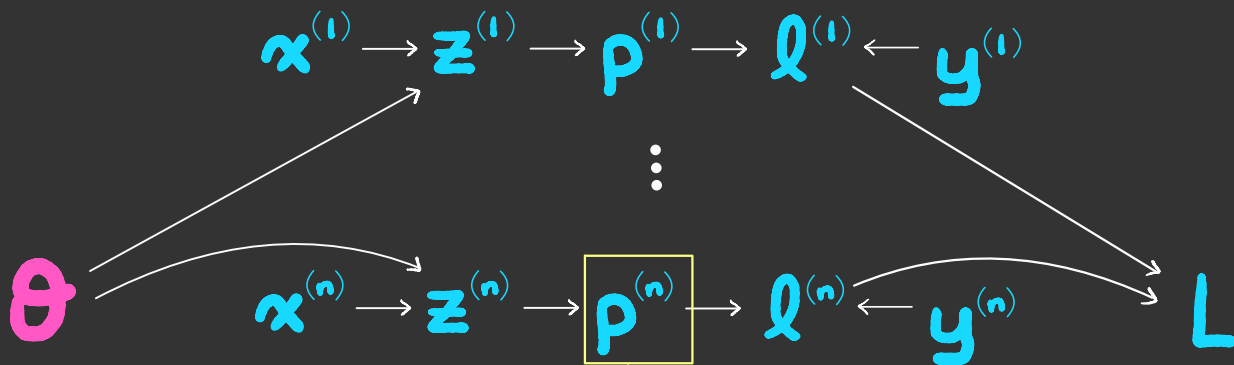
$$\frac{\partial L}{\partial \theta} = \sum_{n=1}^N \frac{1 - 2y^{(n)}}{p^{(n)}y^{(n)} + (1-p^{(n)})(1-y^{(n)})} \frac{\partial p^{(n)}}{\partial z^{(n)}} \frac{\partial z^{(n)}}{\partial \theta}$$



$$\frac{\partial L}{\partial \theta} = \sum_{n=1}^N \frac{1 - 2y^{(n)}}{p^{(n)}y^{(n)} + (1-p^{(n)})(1-y^{(n)})} \boxed{\frac{\partial p^{(n)}}{\partial z^{(n)}}} \frac{\partial z^{(n)}}{\partial \theta}$$



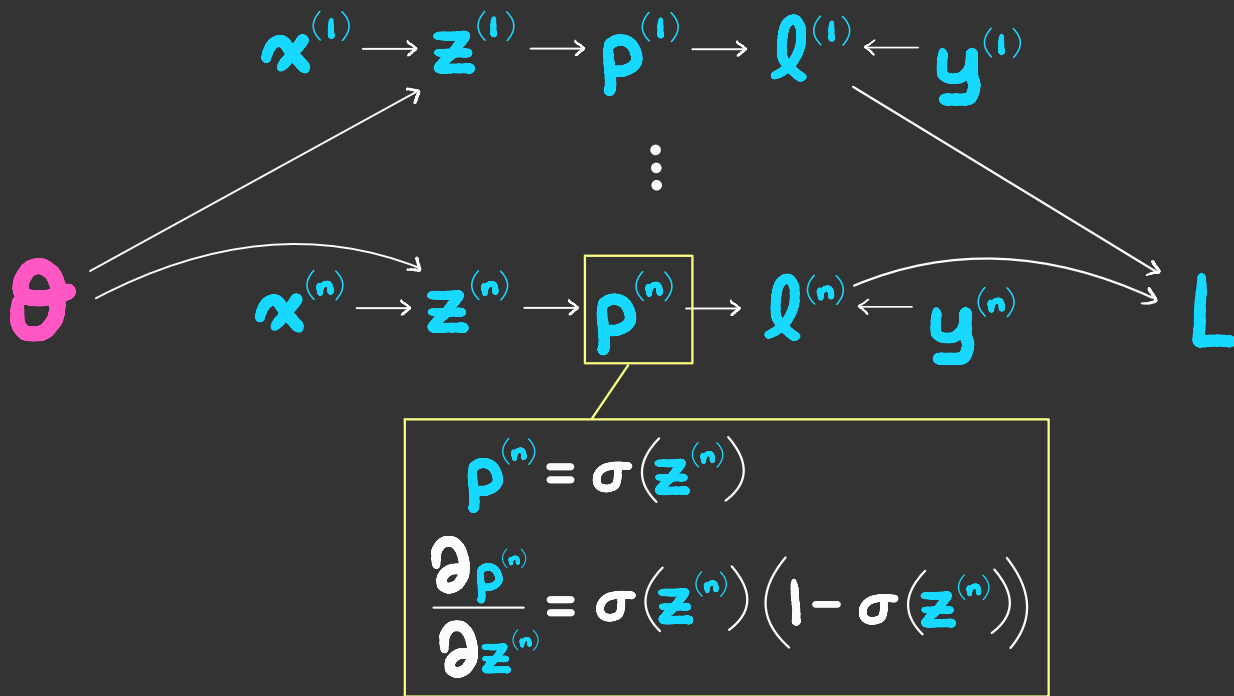
$$\frac{\partial L}{\partial \theta} = \sum_{n=1}^N \frac{1 - 2y^{(n)}}{p^{(n)}y^{(n)} + (1-p^{(n)})(1-y^{(n)})} \boxed{\frac{\partial p^{(n)}}{\partial z^{(n)}}} \frac{\partial z^{(n)}}{\partial \theta}$$



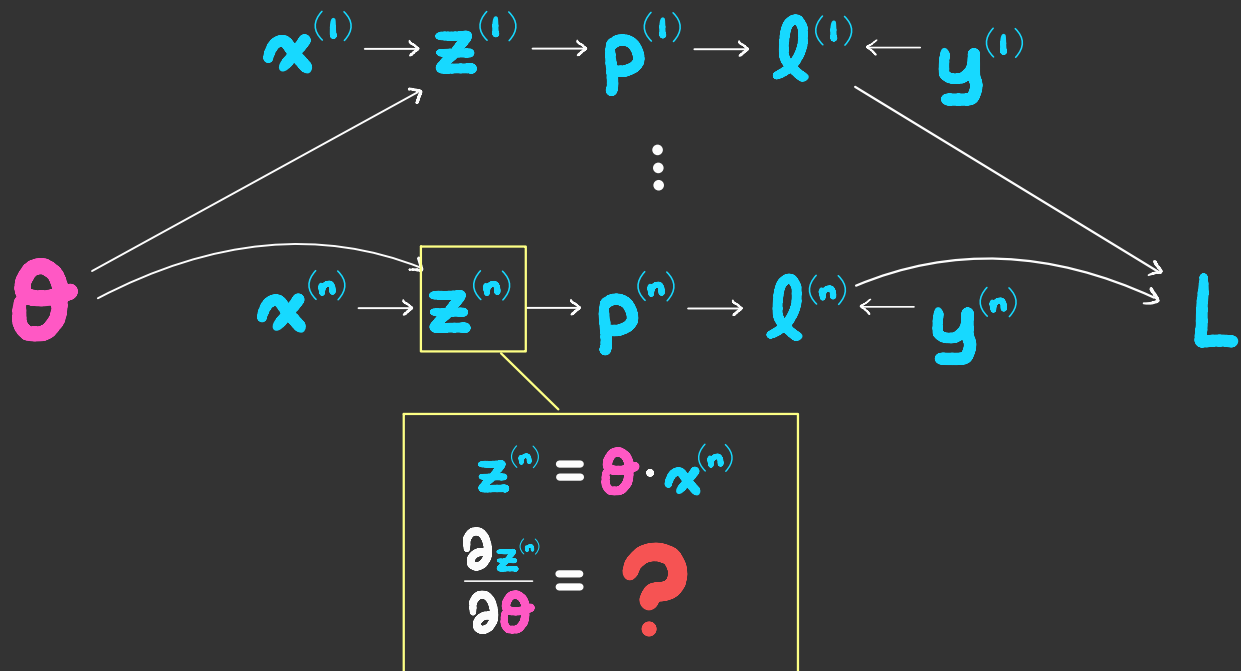
$$p^{(n)} = \sigma(z^{(n)})$$

$$\frac{\partial p^{(n)}}{\partial z^{(n)}} = \sigma(z^{(n)}) (1 - \sigma(z^{(n)}))$$

$$\frac{\partial L}{\partial \theta} = \sum_{n=1}^N \frac{1 - 2y^{(n)}}{p^{(n)}y^{(n)} + (1-p^{(n)})(1-y^{(n)})} \boxed{\sigma(z^{(n)}) (1 - \sigma(z^{(n)}))} \frac{\partial z^{(n)}}{\partial \theta}$$



$$\frac{\partial L}{\partial \theta} = \sum_{n=1}^N \frac{1 - 2y^{(n)}}{p^{(n)}y^{(n)} + (1-p^{(n)})(1-y^{(n)})} \sigma(z^{(n)}) (1 - \sigma(z^{(n)})) \boxed{\frac{\partial z^{(n)}}{\partial \theta}}$$



$$\frac{\partial L}{\partial \theta} = \sum_{n=1}^N \frac{1 - 2y^{(n)}}{p^{(n)}y^{(n)} + (1-p^{(n)})(1-y^{(n)})} \sigma(z^{(n)}) (1 - \sigma(z^{(n)})) \frac{\partial z^{(n)}}{\partial \theta}$$

$$\begin{bmatrix} \frac{\partial L}{\partial \theta_1} \\ \vdots \\ \frac{\partial L}{\partial \theta_D} \end{bmatrix}$$

$$\begin{bmatrix} \theta_1 \\ \vdots \\ \theta_D \end{bmatrix}$$

$$x^{(1)} \rightarrow z^{(1)} \rightarrow p^{(1)} \rightarrow l^{(1)} \leftarrow y^{(1)}$$

$\vdots$

$$\begin{bmatrix} x_1^{(n)} \\ \vdots \\ x_D^{(n)} \end{bmatrix}$$

$$z^{(n)}$$

$$p^{(n)}$$

$$l^{(n)}$$

$$y^{(n)}$$

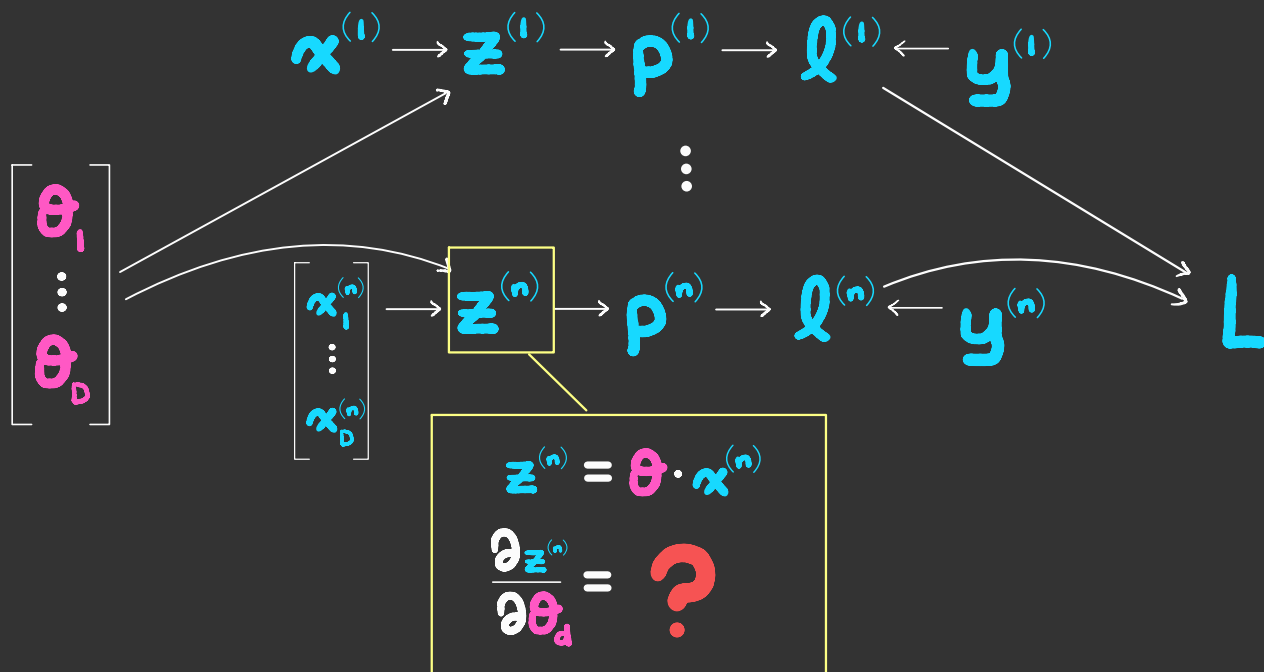
$$L$$

$$\begin{bmatrix} \frac{\partial z^{(n)}}{\partial \theta_1} \\ \vdots \\ \frac{\partial z^{(n)}}{\partial \theta_D} \end{bmatrix}$$

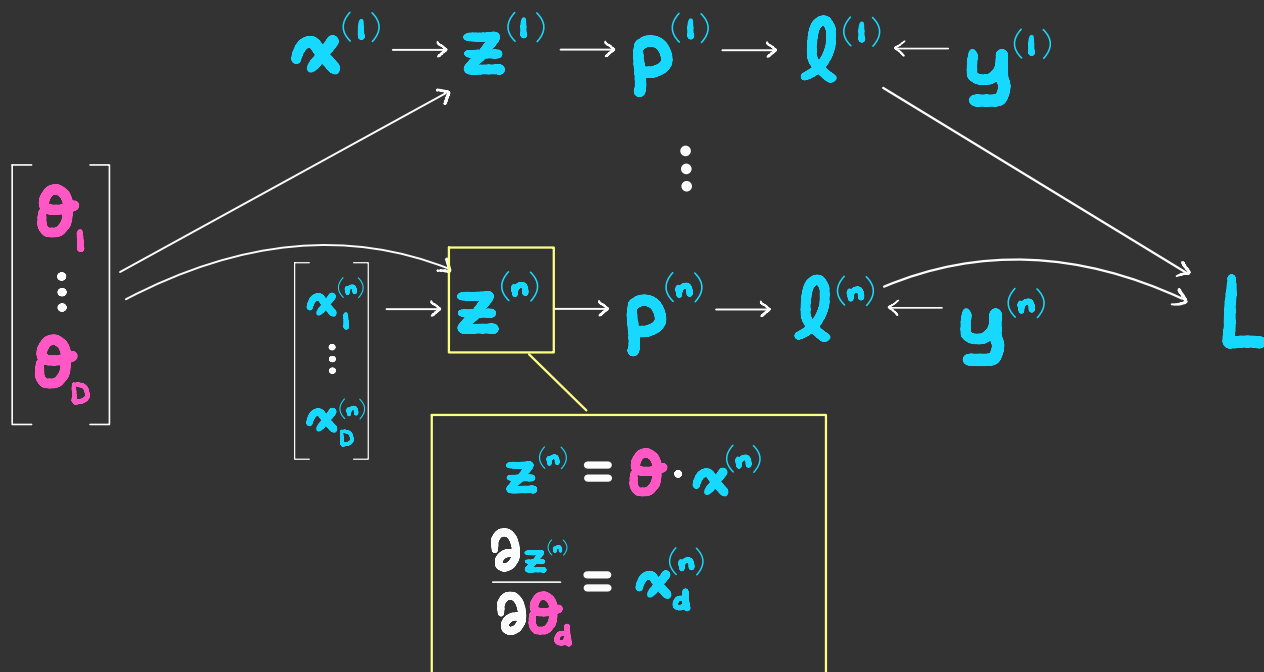
$$z^{(n)} = \theta \cdot x^{(n)}$$

$$\frac{\partial z^{(n)}}{\partial \theta} = ?$$

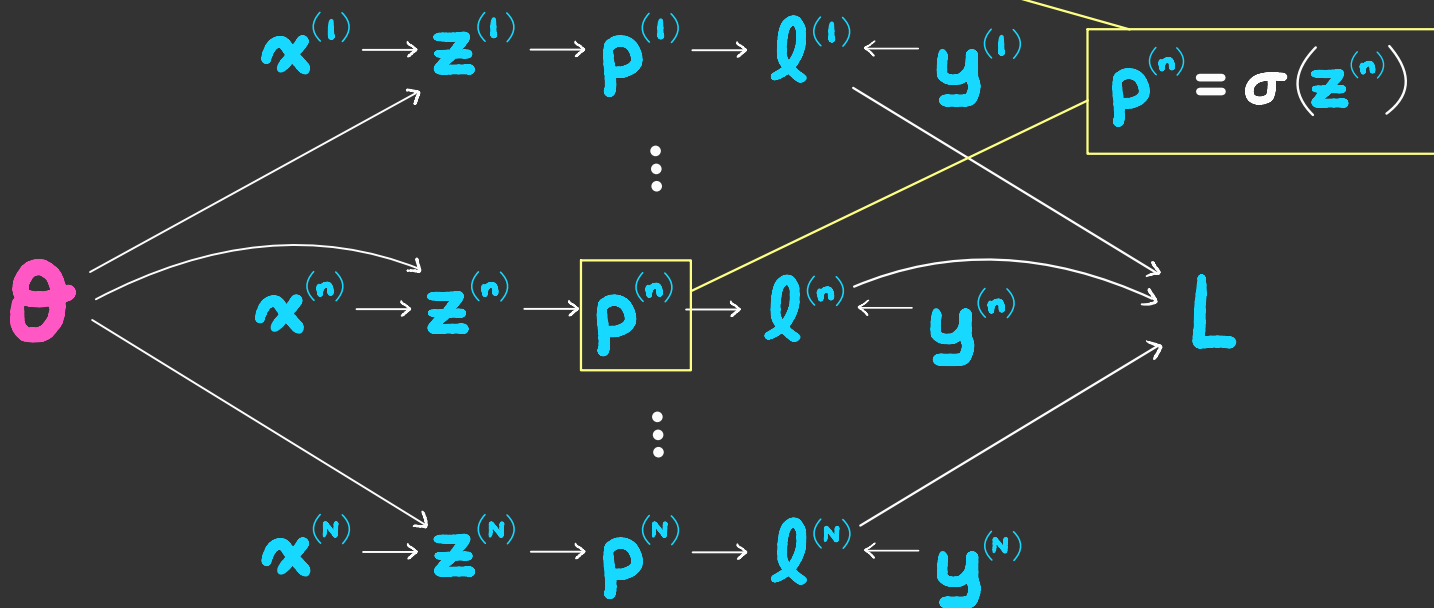
$$\frac{\partial L}{\partial \theta_d} = \sum_{n=1}^N \frac{1 - 2y^{(n)}}{p^{(n)}y^{(n)} + (1-p^{(n)})(1-y^{(n)})} \sigma(z^{(n)}) (1 - \sigma(z^{(n)})) \boxed{\frac{\partial z^{(n)}}{\partial \theta_d}}$$



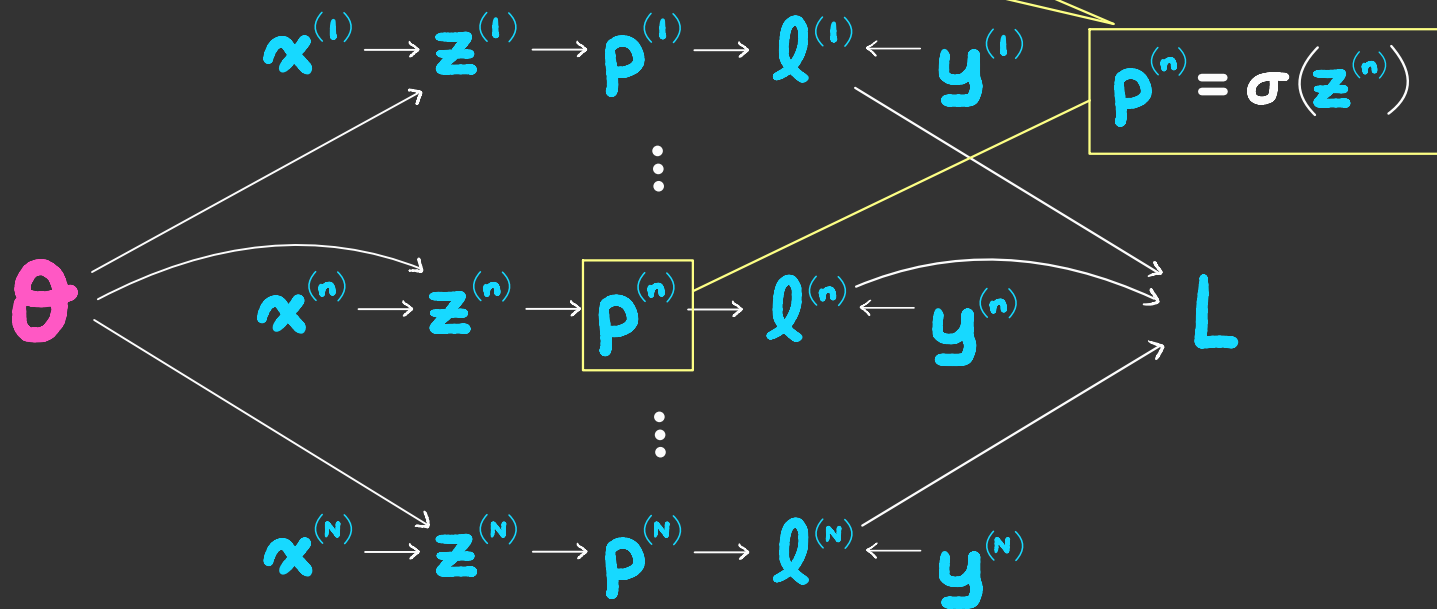
$$\frac{\partial L}{\partial \theta_d} = \sum_{n=1}^N \frac{1 - 2y^{(n)}}{p^{(n)}y^{(n)} + (1-p^{(n)})(1-y^{(n)})} \sigma(z^{(n)}) (1 - \sigma(z^{(n)})) \quad \boxed{x_d^{(n)}}$$



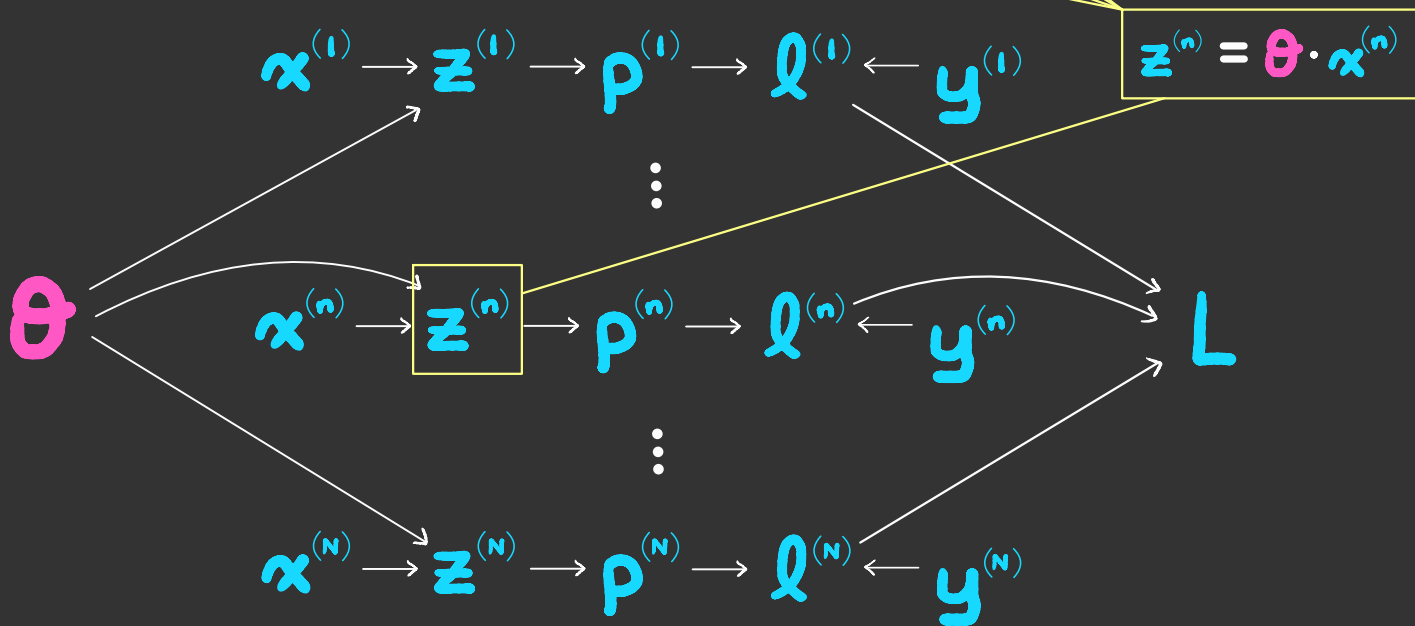
$$\frac{\partial L}{\partial \theta_d} = \sum_{n=1}^N \frac{(1 - 2y^{(n)}) \sigma(z^{(n)}) (1 - \sigma(z^{(n)})) x_d^{(n)}}{\boxed{p^{(n)}} y^{(n)} + (1 - \boxed{p^{(n)}}) (1 - y^{(n)})}$$



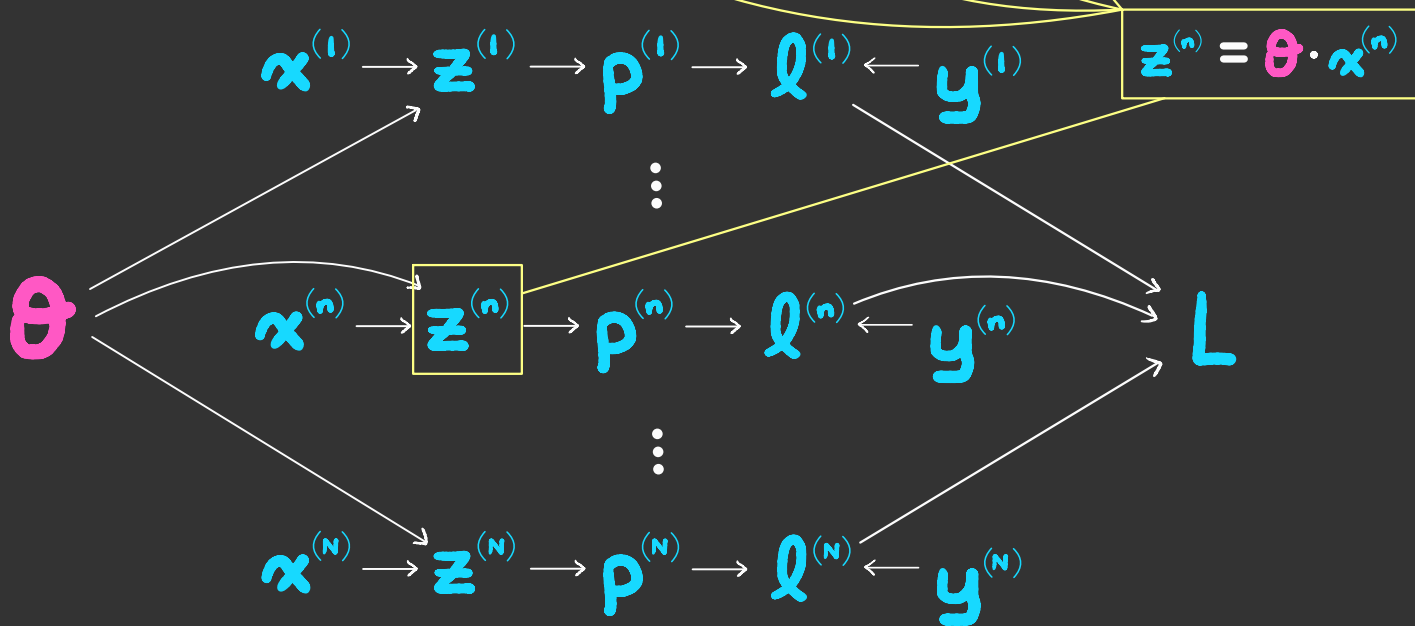
$$\frac{\partial L}{\partial \theta_d} = \sum_{n=1}^N \frac{(1 - 2y^{(n)}) \sigma(z^{(n)}) (1 - \sigma(z^{(n)})) x_d^{(n)}}{\sigma(z^{(n)}) y^{(n)} + (1 - \sigma(z^{(n)})) (1 - y^{(n)})}$$



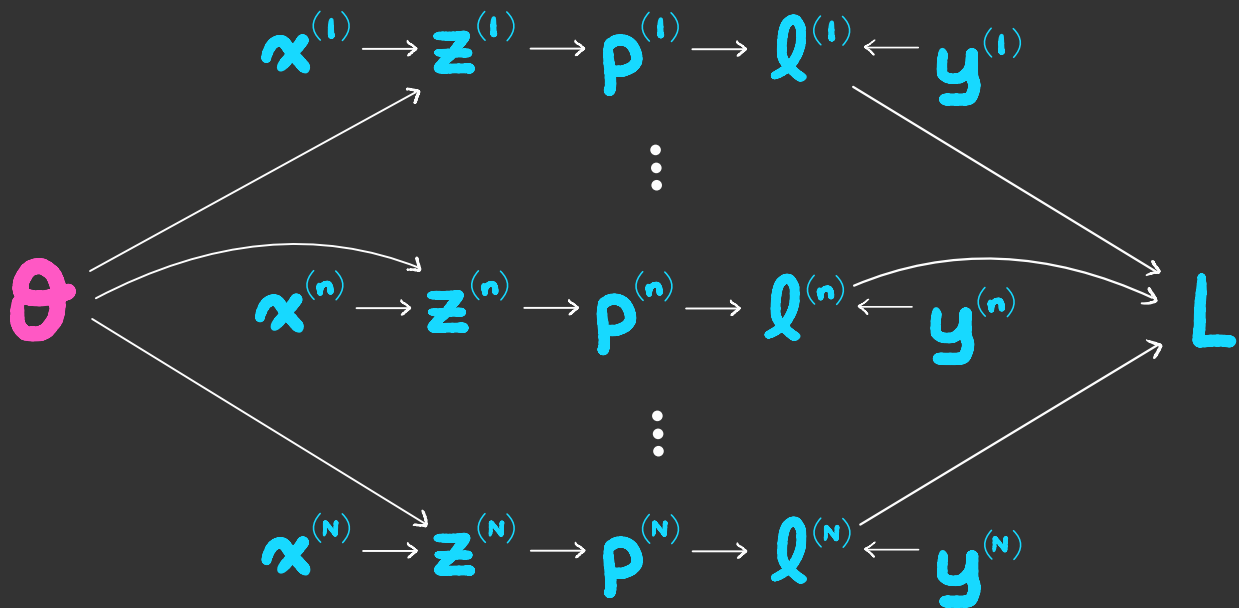
$$\frac{\partial L}{\partial \theta_d} = \sum_{n=1}^N \frac{(1 - 2y^{(n)}) \sigma(z^{(n)}) (1 - \sigma(z^{(n)})) x_d^{(n)}}{\sigma(z^{(n)}) y^{(n)} + (1 - \sigma(z^{(n)})) (1 - y^{(n)})}$$



$$\frac{\partial L}{\partial \theta_d} = \sum_{n=1}^N \frac{(1 - 2y^{(n)}) \sigma(\theta \cdot x^{(n)}) (1 - \sigma(\theta \cdot x^{(n)})) x_d^{(n)}}{\sigma(\theta \cdot x^{(n)}) y^{(n)} + (1 - \sigma(\theta \cdot x^{(n)})) (1 - y^{(n)})}$$



$$\frac{\partial L}{\partial \theta_d} = \sum_{n=1}^N \frac{(1 - 2y^{(n)}) \sigma(\theta \cdot x^{(n)}) (1 - \sigma(\theta \cdot x^{(n)})) x_d^{(n)}}{\sigma(\theta \cdot x^{(n)}) y^{(n)} + (1 - \sigma(\theta \cdot x^{(n)})) (1 - y^{(n)})}$$



logistic regression

model space

$$y = \left(\begin{bmatrix} \theta_1 \\ \vdots \\ \theta_D \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_D \end{bmatrix} > 0 \right)$$

training data

		instances		
		1	...	N
features	x_1	$x_1^{(1)}$	...	$x_1^{(N)}$
	$\vdots$	$\vdots$		$\vdots$
	x_D	$x_D^{(1)}$	...	$x_D^{(N)}$
outcome	y	$y^{(1)}$	...	$y^{(N)}$
		binary values		

loss function

$$L(\theta) = -\sum_{n=1}^N \log(p^{(n)} y^{(n)} + (1-p^{(n)})(1-y^{(n)}))$$

$$\text{where } p^{(n)} = \sigma \left(\begin{bmatrix} \theta_1 \\ \vdots \\ \theta_D \end{bmatrix} \cdot \begin{bmatrix} x_1^{(n)} \\ \vdots \\ x_D^{(n)} \end{bmatrix} \right)$$

optimization

$$\underset{\theta}{\operatorname{argmin}} L(\theta)$$

gradient

$$\frac{\partial L}{\partial \theta} = \begin{bmatrix} \frac{\partial L}{\partial \theta_1} \\ \vdots \\ \frac{\partial L}{\partial \theta_D} \end{bmatrix}$$

